



Wayanamac Education Trust®
DON BOSCO INSTITUTE OF TECHNOLOGY
Kumbalagodu, Mysore road, Bangalore - 560074



Affiliated
to VTU
Belagavi



Approved by
AICTE, New
Delhi



Accredited by
NAAC,
Bengaluru



Accredited
by NBA*,
New Delhi



Recognized
by Govt. of
Karnataka



Department of Information Science and Engineering
Academic year: 2025-26
Mini Project– Review Presentation
On

SMART DEPARTMENT PORTAL

Name of Student:

Adithya K(1DB23IS007)

Jaishree K S(1DB23IS064)

GUIDE:

Mr. Kiran Kumar D

Assistant Professor

Department of IS&E

DBIT, Bengaluru

Contents

- Introduction
- Problem Statement
- Motivation
- Aim & Objectives
- System Design
- Methodologies
- Results and snapshots
- Application of the project
- Conclusion
- Future Scope of the Work
- References

Introduction

- A Smart Department Portal (ISEHUB) is a comprehensive software application, often web-based, designed to centralize and streamline the administrative, academic, and operational tasks of an academic or organizational department.
- It serves as a digital hub for managing student and staff information, academic records, attendance, event details, and communication, ultimately aiming to reduce manual paperwork, improve efficiency, and provide real-time access to critical data for all users, including administrators, faculty, and students.
- ❑ **Faculty:** Access student records, manage course-related information.
- ❑ **Students:** View their academic records, attendance, schedules, and receive important announcements.
- ❑ **Admins:** Manage system settings, user access, and oversee department operations.

Problem Statement

In most college departments, key activities such as student information management, notice updates, attendance tracking, and communication between faculty and students are handled manually .

This leads to:

- Inefficient communication and information delays
- Difficulty in accessing academic records
- Lack of a centralized system for managing departmental tasks
- Increased chances of data loss or duplication

To overcome these challenges, there is a need for a Smart Department Portal — a centralized, web-based system that streamlines communication, automates routine processes, and provides quick access to important departmental information for students, faculty, and administrators.

Motivation

- Departmental activities such as student data management, announcements, and attendance tracking are often handled through multiple manual processes or unconnected systems.
- This causes issues like data redundancy, delays in communication, and lack of centralized access.
- With increasing digitization in education, there is a need for a **smart, integrated web portal** to streamline these operations.
- The Smart Department Portal aims to provide an efficient and secure platform for managing departmental data, improving accessibility, and promoting digital transformation within the institution.

Aim and Objectives

The main aim of the Smart Department Portal is to design and develop a centralized, web-based system that automates and simplifies the management of departmental activities such as student information, announcements, attendance, and internal assessments.

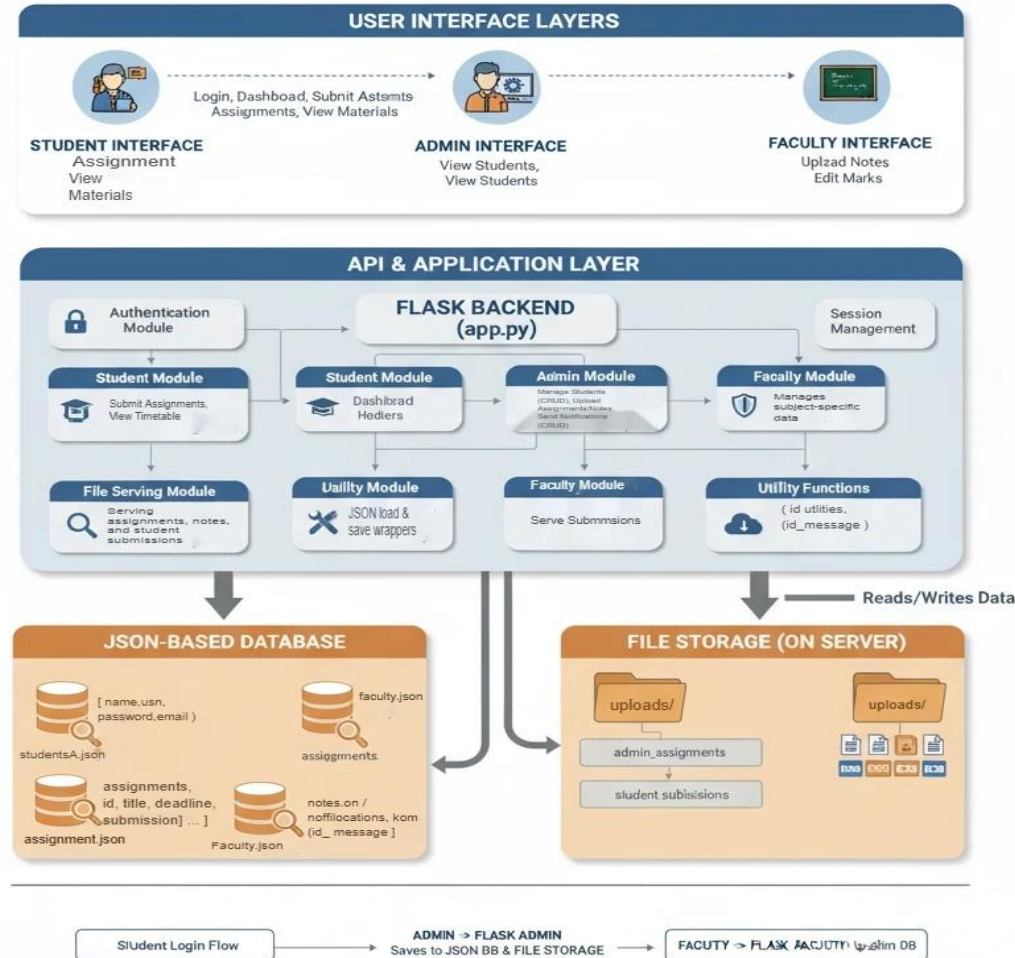
It aims to enhance communication, efficiency, and transparency between students, faculty, and administrators

Objectives:

- To promote a **digital and paperless environment** within the department.
- To facilitate **seamless communication** among students, faculty
- To design and develop a **web-based centralized portal** for managing departmental operations efficiently.
- To **automate departmental** processes such as announcements, attendance, and internal assessments.
- To maintain a **centralized database** for storing and retrieving student and faculty information.

System Design

ISEHUB SYSTEM ARCHITECTURE



Methodologies

1. Requirement Analysis

- Identify user roles: **Faculty, Students, Admin**
- Gather requirements for academic, communication, attendance, and assignment modules
- Prioritize features based on usability and department workflow

2. System Planning

- Break project into modules:
 - Academic
 - Communication
 - Attendance
 - Assignments
 - Notifications

Results and snapshots

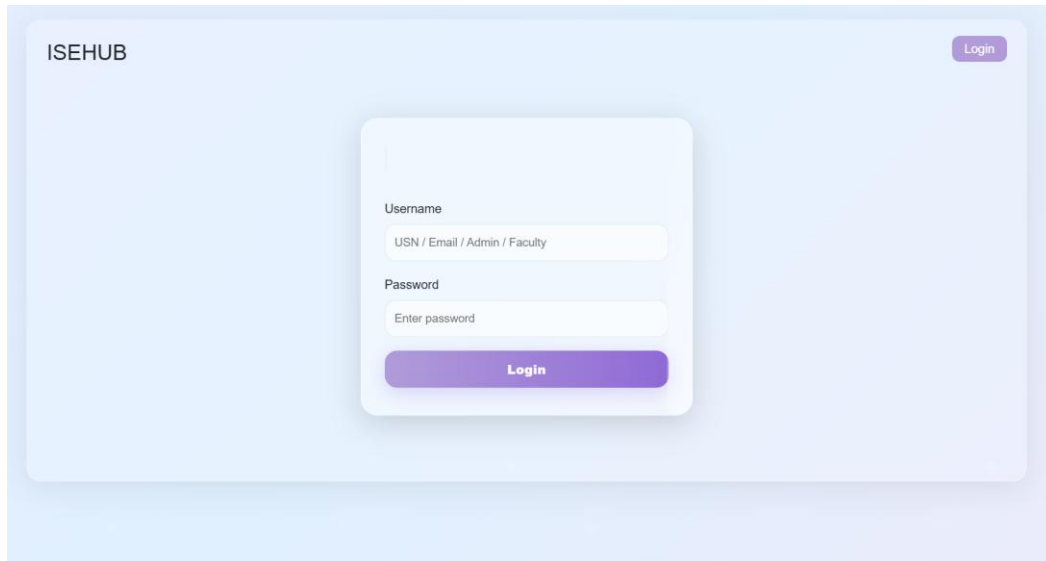


Fig1.Login Page

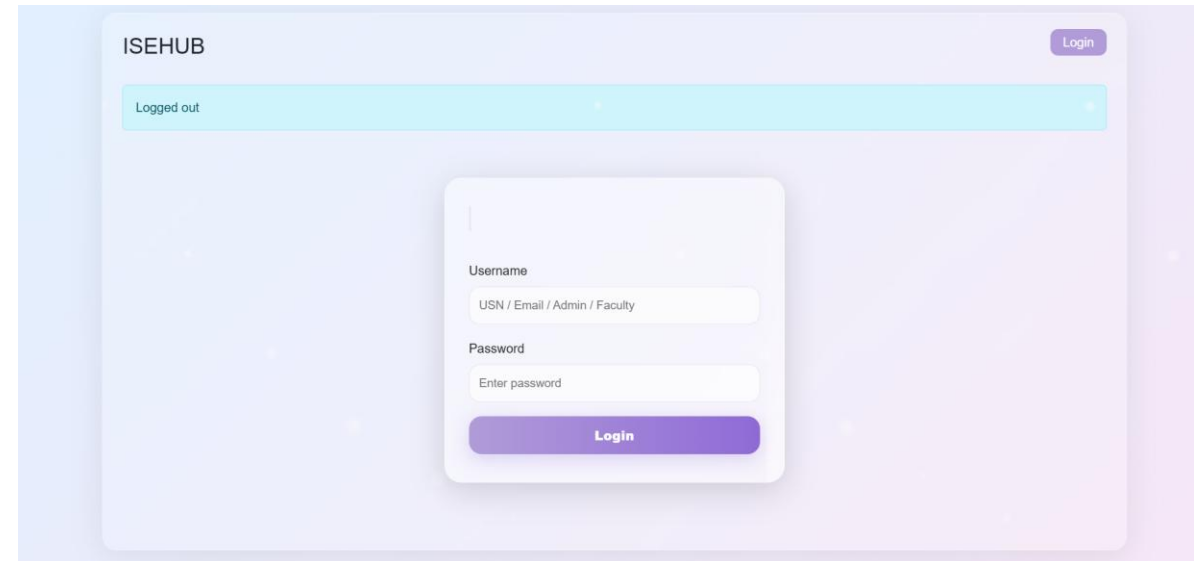


Fig2.Logout Page

ADMIN DASHBOARD

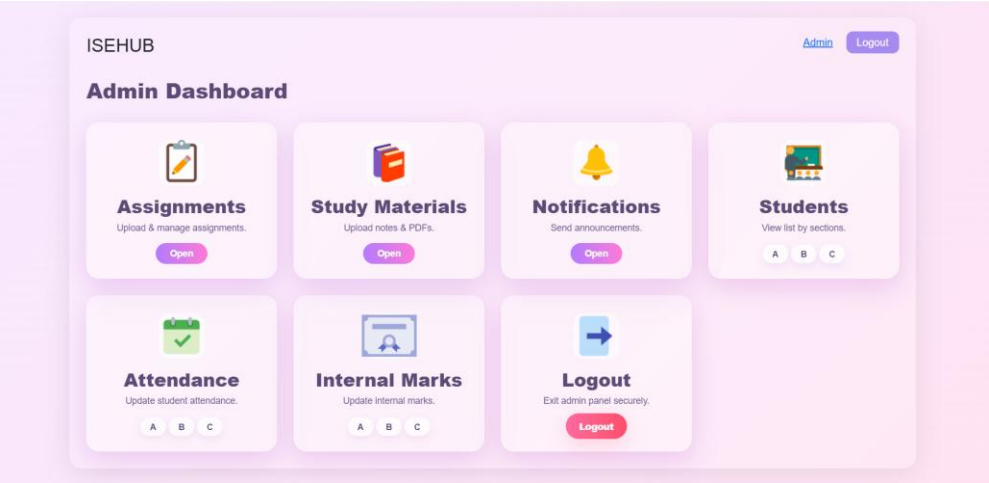


Fig3.Admin Dashboard

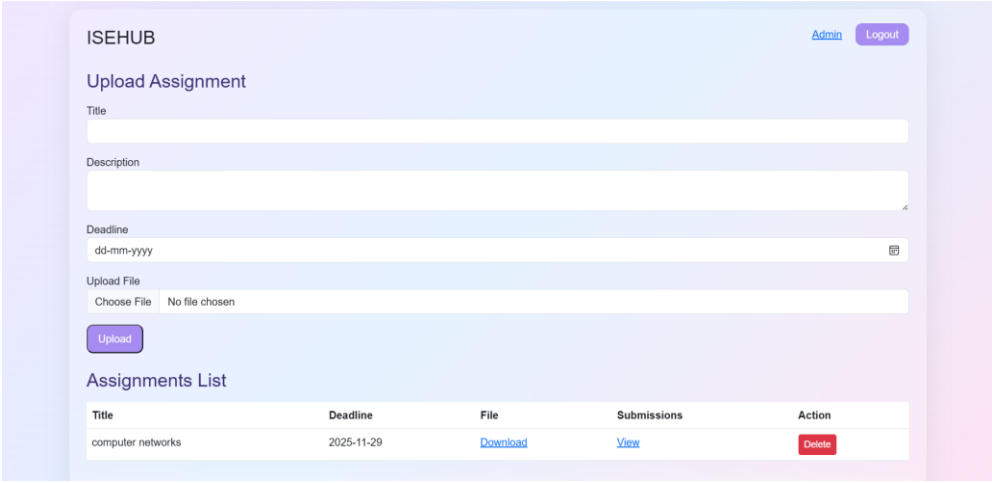


Fig4.Upload Assignments



Fig5.Upload Notes

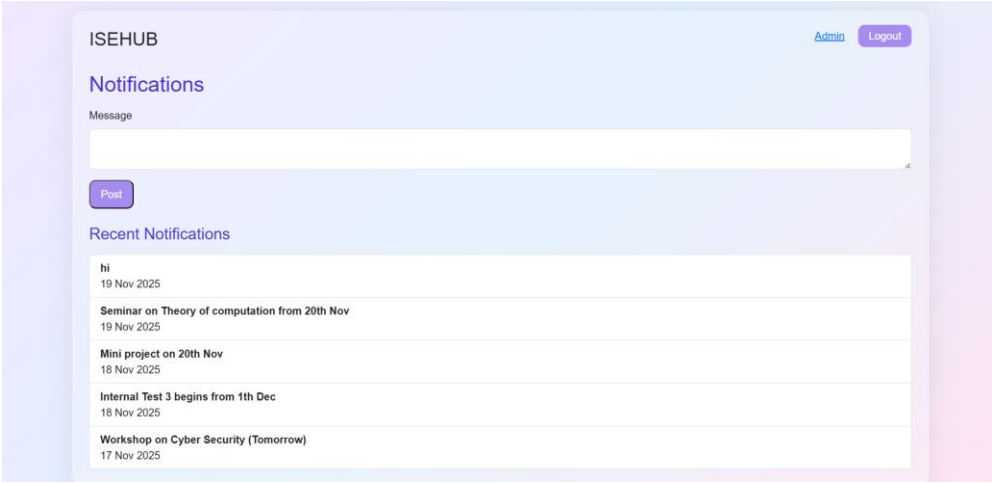


Fig6.Notifications

ISEHUB

[Admin](#)[Logout](#)

Students — Section A

#	Name	USN	Email	Attendance	Actions
1	A L ABHISHEK	1DB23IS001	1DB23IS001@isehub.edu	85%	Attendance Marks
2	ABHISHEK K C	1DB23IS004	1DB23IS004@isehub.edu	79%	Attendance Marks
3	ADITHYA K	1DB23IS007	1DB23IS007@isehub.edu	83%	Attendance Marks
4	AFREED AHMED	1DB23IS009	1DB23IS009@isehub.edu	70%	Attendance Marks
5	AMOGH TEEKA	1DB23IS010	1DB23IS010@isehub.edu	92%	Attendance Marks
6	AMRUTHA G	1DB23IS012	1DB23IS012@isehub.edu	84%	Attendance Marks
7	ANJALI	1DB23IS013	1DB23IS013@isehub.edu	68%	Attendance Marks
8	ASHOK GOWDA S P	1DB23IS014	1DB23IS014@isehub.edu	72%	Attendance Marks
9	B ANAND	1DB23IS015	1DB23IS015@isehub.edu	63%	Attendance Marks
10	BHOOMIKA K	1DB23IS016	1DB23IS016@isehub.edu	91%	Attendance Marks
11	BHOYI GAUTAMI RAVI	1DB23IS017	1DB23IS017@isehub.edu	77%	Attendance Marks
12	BINDU V GOWDA	1DB23IS018	1DB23IS018@isehub.edu	66%	Attendance Marks
13	CHAITANYA R C	1DB23IS019	1DB23IS019@isehub.edu	89%	Attendance Marks

Fig7.Students

ISEHUB

[Admin](#)[Logout](#)

Attendance — Section A

Click Edit to update attendance for that student.

[Back to Section Students](#)

A L ABHISHEK

USN: 1DB23IS001 •
1DB23IS001@isehub.edu

Attendance: 85%

Edit Attendance

ABHISHEK K C

USN: 1DB23IS004 •
1DB23IS004@isehub.edu

Attendance: 79%

Edit Attendance

ADITHYA K

USN: 1DB23IS007 •
1DB23IS007@isehub.edu

Attendance: 83%

Edit Attendance

AFREED AHMED

USN: 1DB23IS009 •
1DB23IS009@isehub.edu

Attendance: 70%

Edit Attendance

AMOGH TEEKA

USN: 1DB23IS010 •
1DB23IS010@isehub.edu

Attendance: 92%

Edit Attendance

AMRUTHA G

USN: 1DB23IS012 •
1DB23IS012@isehub.edu

Attendance: 84%

Edit Attendance

ANJALI

USN: 1DB23IS013 •
1DB23IS013@isehub.edu

Attendance: 68%

Edit Attendance

ASHOK GOWDA S P

USN: 1DB23IS014 •
1DB23IS014@isehub.edu

Attendance: 72%

Edit Attendance

Fig8.Attendance

ISEHUB

[Admin](#)[Logout](#)

Edit Attendance — AFIYA ARFAIN

USN: 1DB23IS008 | Section: B

Attendance (%)

88

Save

Fig9.Edit Attendance

Update Internal Marks — ABHISHEK C

USN: 1DB23IS002 • Section: C

SEPM

17

CN

13

TOC

19

AI

14

RMIPR

18

EVS

16

Save Marks

Fig10.Update Internal Marks

FACULTY DASHBOARD

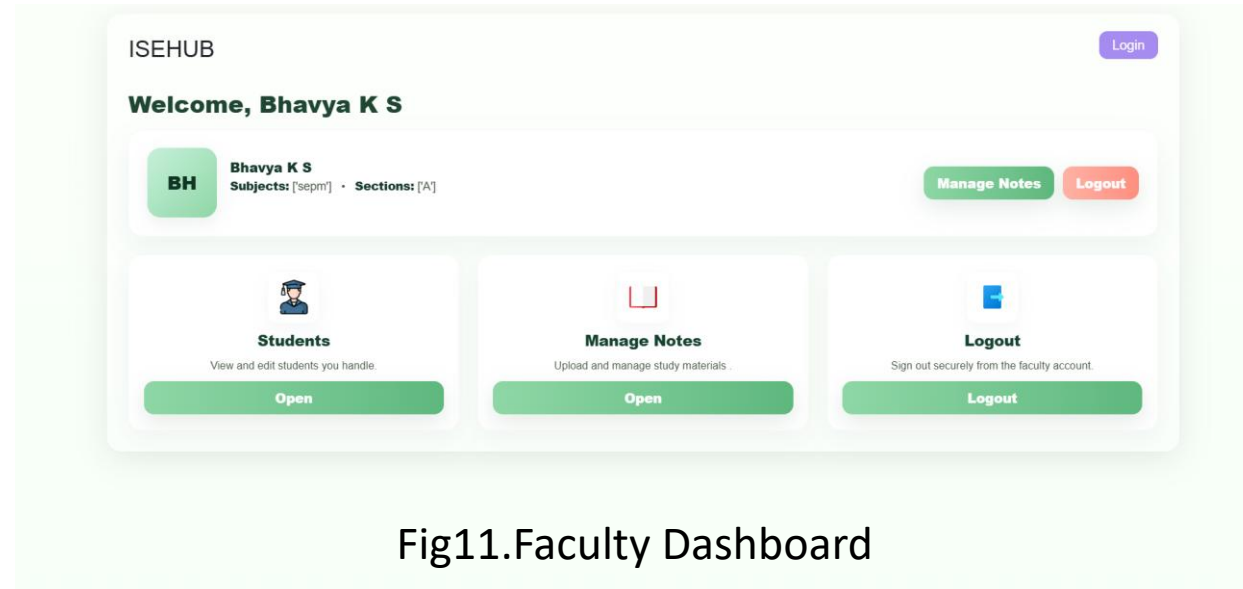
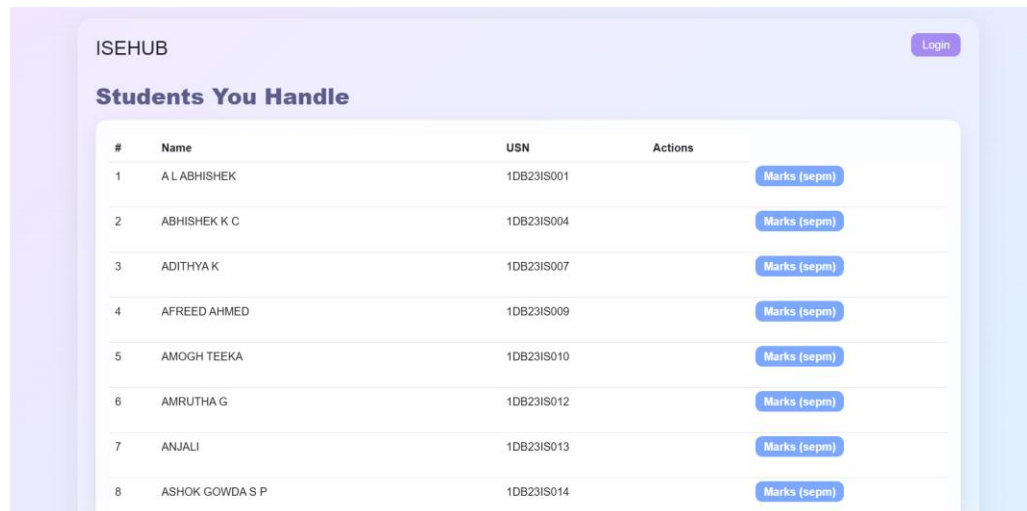


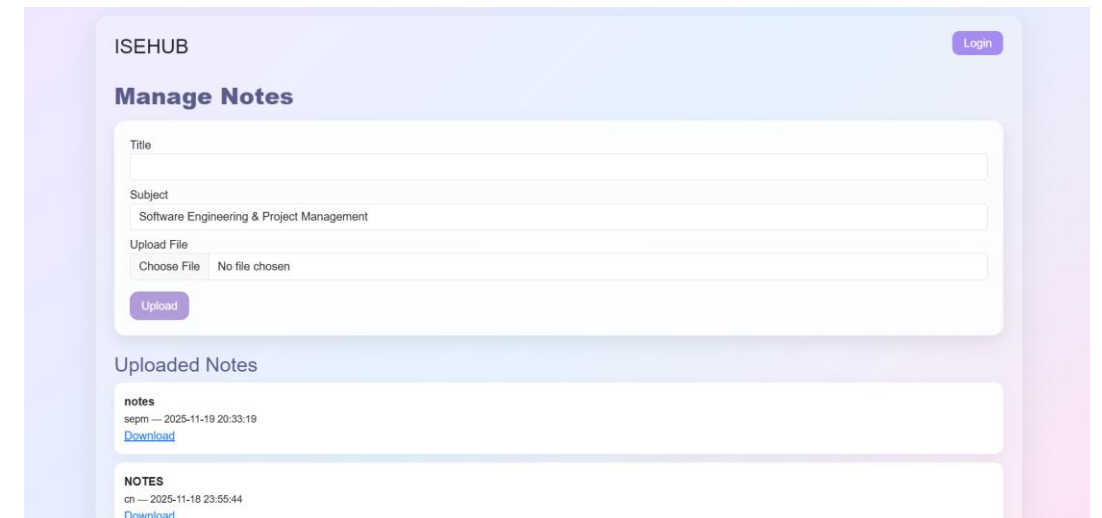
Fig11.Faculty Dashboard



The 'Students You Handle' section (Fig12) displays a table of students managed by the faculty member. The table has columns for #, Name, USN, and Actions. Each row represents a student, and the 'Actions' column contains a 'Marks (sepm)' button.

#	Name	USN	Actions
1	A L ABHISHEK	1DB23IS001	Marks (sepm)
2	ABHISHEK K C	1DB23IS004	Marks (sepm)
3	ADITHYA K	1DB23IS007	Marks (sepm)
4	AFREED AHMED	1DB23IS009	Marks (sepm)
5	AMOGH TEEKA	1DB23IS010	Marks (sepm)
6	AMRUTHA G	1DB23IS012	Marks (sepm)
7	ANJALI	1DB23IS013	Marks (sepm)
8	ASHOK GOWDA S P	1DB23IS014	Marks (sepm)

Fig12.Students



The 'Manage Notes' section (Fig13) allows the faculty member to upload and manage notes. It includes a form for adding a new note with fields for Title, Subject, and Upload File. Below the form is a section for 'Uploaded Notes' showing a list of notes with their titles, subjects, and upload dates. Each note has a 'Download' button.

ISEHUB

Manage Notes

Title

Subject

Software Engineering & Project Management

Upload File

Choose File No file chosen

Upload

Uploaded Notes

notes

sepm — 2025-11-19 20:33:19

Download

NOTES

cn — 2025-11-18 23:55:44

Download

Fig13.Manage Notes

STUDENT DASHBOARD

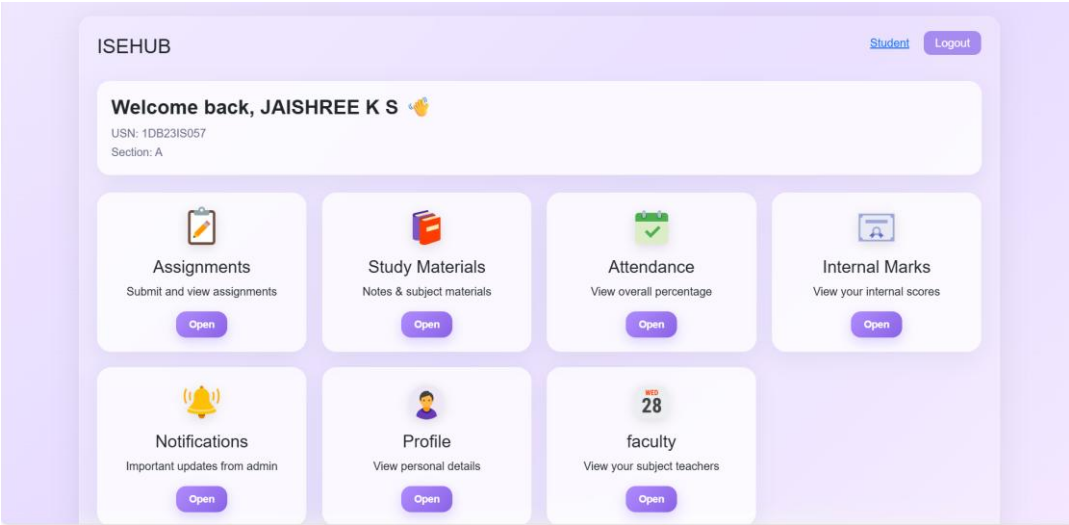


Fig14.Student Dashboard

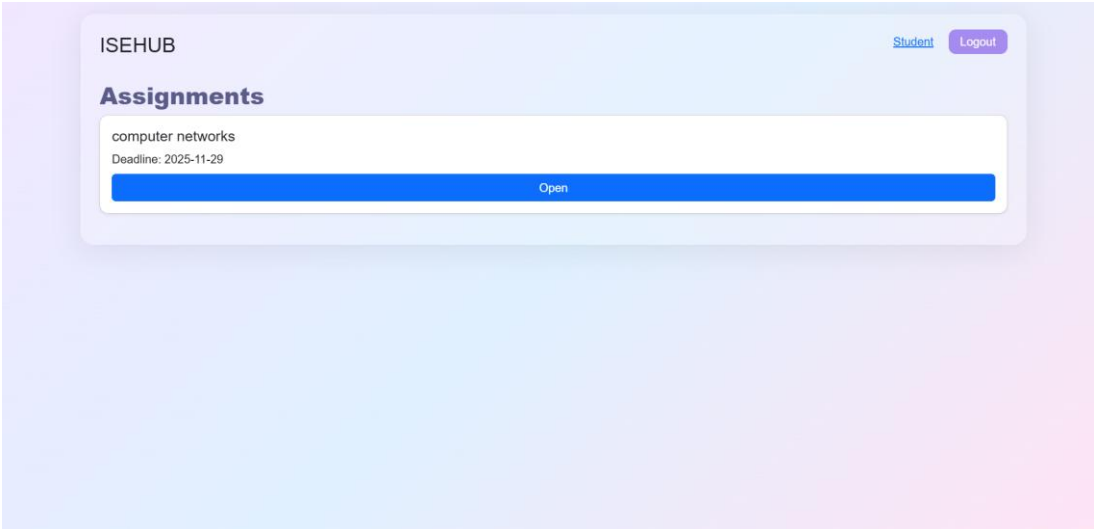


Fig15.Assignments

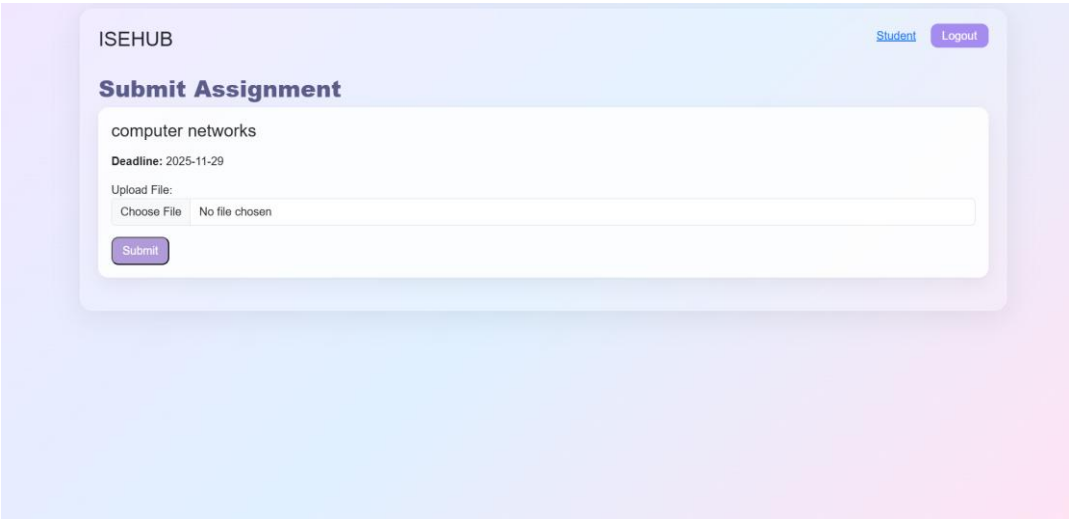


Fig16.Submit Assignment

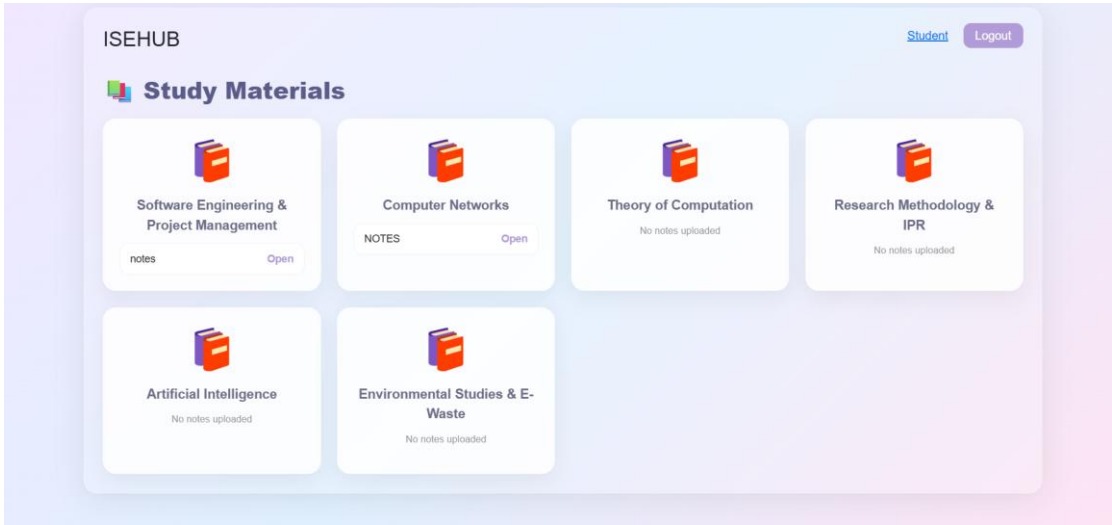


Fig17.Study Materials

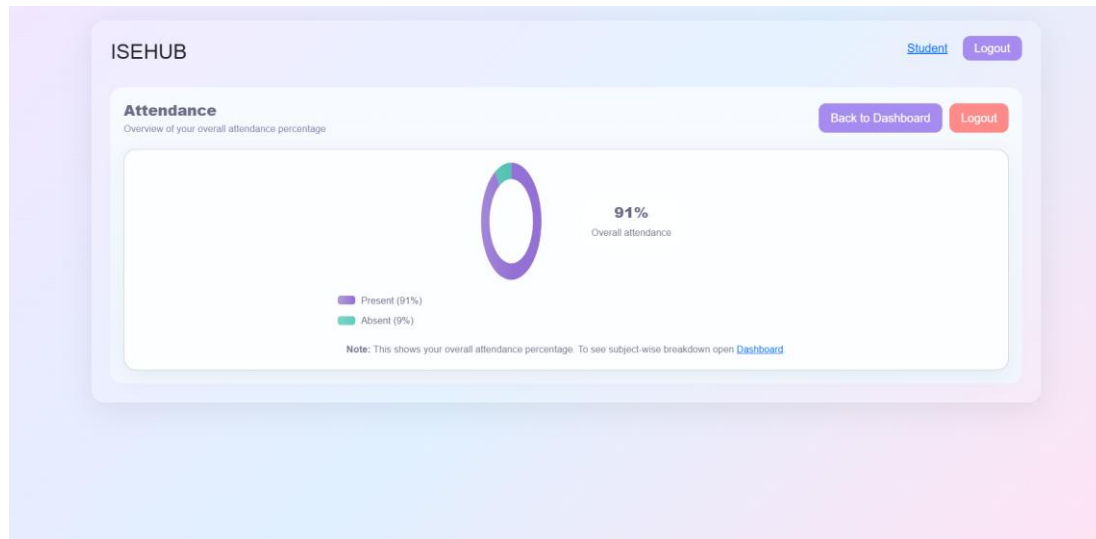


Fig18.Attendance

The table lists internal marks for different subjects. Each row has a subject name on the left and a score in a purple pill-shaped box on the right.

Subject	Score
SEPM	20
CN	19
TOC	18
AI	20
EVS	20
RM/IPR	19

Fig19.Internal Marks

The notifications list contains five items, each with a title and a date.

Notification	Date
hi	19 Nov 2025
Seminar on Theory of computation from 20th Nov	19 Nov 2025
Mini project on 20th Nov	18 Nov 2025
Internal Test 3 begins from 1th Dec	18 Nov 2025
Workshop on Cyber Security (Tomorrow)	17 Nov 2025

Fig20.Notifications

The profile page displays student information and performance metrics.

Section	Details
Profile	JAISHREE K S USN: 1DB23IS057 Section: A
Basic Details	Email: 1DB23IS057@isehub.edu Phone: 9901456723
Attendance	Overall Attendance: 91%
Internal Marks	SEPM: 20 / 20 CN: 19 / 20 TOC: 18 / 20

Fig21.Profile

Application of the Project

The Smart Department Portal can be effectively utilized across educational institutions for digital management of departmental operations. It enables seamless coordination among students, faculty by providing a unified platform for communication, record maintenance, and information sharing. Faculty can manage attendance, internal marks, and announcements efficiently, while students can access notices and academic updates in real-time. Department administrators can use the system to store and retrieve data, generate reports, and monitor overall performance.

Furthermore, the system can be scaled to function at an institutional level, integrating multiple departments into a centralized digital ecosystem that promotes automation, transparency, and paperless administration.

Conclusion

- The Smart Department Portal effectively streamlines departmental activities by integrating communication, data management, and automation into a single digital platform.
- It reduces manual effort, enhances transparency, and promotes efficient collaboration between students, faculty
- Overall, the system contributes to building a smarter and more organized department environment.

Future Enhancements:

- Integration of Mobile application support for better accessibility.
- Implementation of AI-based notifications and chatbots for instant updates.
- Addition of data analytics to track academic performance and departmental trends.
- Integration with College-wide systems for institution-level management.
- Use of Cloud storage for scalability and data security.

Future Scope of the work

1. AI-Powered Analytics

- Predict student performance
- Automated attendance predictions

2. Advanced Security

- End-to-end encrypted messages
- Multi-factor authentication

3. Cloud Scalability

- Auto-scaling servers for large institutions

4. Biometric + RFID Attendance

- Face recognition attendance
- RFID-based in & out tracking

References

- 1] S. Pavithra *et al.*, “Department Information System – Web Application,” *IJERT*, vol. 8, no. 5, 2019.
- [2] I. Z. Kogotkova *et al.*, “Smart Department: Designing Digital Applications for Smart Organizations,” *Springer*, 2021.
- [3] F. M. Gonzalez- Longatt *et al.*, “Secured Student Portal Using Cloud,” *Expert Clouds and Applications*, 2021.
- [4] A. C. Doctor, “Integrated Educational Management Tool for Adamson University,” *arXiv preprint arXiv:2212.08039*, 2022.
- [5] “Web Portals for Higher Education Institutions: A Review,” *Int. J. of Information Management*, 2018.
- [6] GitHub, “Open-Source Department Portal Projects,” Available: <https://github.com>.

THANK YOU